



UNIVERSITÀ DI TRENTO

Department of Information Engineering and Computer Science

Course: Data Mining

Can Musical Genres Be Recovered from Acoustic Features? A Clustering and Network Analysis of Spotify Tracks

Author: Pietro Farina

Student Number: 249889

Academic Year 2025/2026

Contents

Introduction	1
1 Dataset Construction and Preparation	2
1.1 Dataset Description	2
1.2 Data Standardization	2
1.2.1 Genre Harmonization	3
1.3 Dataset Compatibility Assessment	3
1.4 Preliminary Exploratory Data Analysis	4
2 Can Musical Genres Be Recovered from Acoustic Features?	6
2.1 Motivation	6
2.2 Acoustic Feature Representation	6
2.3 Clustering Methodology	7
2.4 Results on Macro Genres	8
2.4.1 Natural Acoustic Partitions	8
2.4.2 Genre-Oriented Partition	9
2.5 Results on Subgenres	10
3 Network Analysis	12
3.1 Motivation	12
3.2 Song Similarity Network	12
3.3 Global Network Properties	12
3.4 Community Detection	13
3.5 Communities versus Genres	13
3.6 Acoustic Characterization of the Detected Communities	13
3.7 Network Structure within Macro-Genres	15
3.8 Structurally Central Songs within Communities	16
4 Discussion	18
4.1 Main Findings	18
4.2 K-Means and Louvain: Complementary Views of Acoustic Structure	18
4.3 Implications for Music Discovery and Recommendation	19
4.3.1 Illustrative Song-to-Song Paths	19
Conclusions	21
Bibliography	22
Appendix	23

Introduction

The rapid growth of music streaming platforms has transformed the way listeners discover, organize, and consume music. Services such as Spotify provide access to millions of tracks accompanied by rich metadata and a set of automatically extracted audio features describing characteristics such as danceability, energy, acousticness, valence, and tempo. These descriptors offer a quantitative representation of the musical content, enabling data-driven analyses of musical similarity and offering an objective basis for classifying sound.

Traditionally, music has been organized through genre labels, which provide a convenient semantic categorization for listeners, artists, and recommendation systems. Genres are widely used to facilitate music discovery, playlist generation, and catalog organization. However, the notion of genre is inherently subjective and continuously evolving, shaped by cultural conventions and the consensus of musicians, critics, and listeners rather than by any fixed physical laws. Contemporary songs frequently blend influences from multiple musical traditions, making genre boundaries increasingly ambiguous. Consequently, songs assigned to the same genre often exhibit substantial acoustic differences, while tracks from distinct genres can converge in their underlying feature space, producing unexpected structural similarities.

This raises an important question: **do musical genres correspond to natural structures in the acoustic feature space, or are they primarily semantic labels that only partially describe the underlying characteristics of music?** In other words, if genre labels were removed entirely, would songs naturally organize into genre-based groups, or would alternative patterns emerge from their acoustic similarities?

To investigate this question, this study analyzes a large collection of Spotify tracks using a combination of unsupervised learning and network science techniques. The analysis begins with data preprocessing and exploratory analysis, followed by clustering methods to evaluate whether traditional genre labels can be recovered solely from audio features. The clustering results are then complemented by a graph-based representation of song similarity, followed by community detection to identify groups of tracks sharing local acoustic characteristics independently of their assigned genres.

1 Dataset Construction and Preparation

1.1 Dataset Description

The analyses presented in this work are based on Spotify tracks described by both semantic metadata and automatically extracted audio features. While metadata such as track title, artist, album and genre provide contextual information, the audio descriptors constitute an objective numerical representation of the musical content, making them particularly suitable for quantitative analyses of similarity and structure.

Rather than relying on a single data source, the final corpus was constructed by integrating two publicly available Spotify datasets: “30,000 Spotify Songs” [3] and the “Spotify Tracks Dataset” [4]. The first dataset contains approximately 32,833 tracks and provides a broad genre assignment for each song. The second dataset contains approximately 114,000 tracks and extends this information by including a substantially larger collection of songs together with a much more detailed genre taxonomy composed of numerous subgenres.

Each track is associated with a set of audio descriptors extracted through Spotify’s audio analysis pipeline. These variables quantify different musical characteristics—including rhythmic, harmonic, dynamic and timbral properties—and form the feature space investigated throughout this study. Table 1.1 summarizes the audio features considered during the analysis.

Table 1.1 Spotify audio features used throughout the study.

Feature	Description
Danceability	Measures how suitable a track is for dancing based on rhythm, tempo and beat consistency.
Energy	Represents the perceived intensity and activity of a track.
Key	Estimated musical key.
Loudness	Overall loudness expressed in decibels (dB).
Mode	Indicates whether the composition is in a major or minor key.
Speechiness	Measures the presence of spoken words.
Acousticness	Confidence measure that a track is acoustic.
Instrumentalness	Estimates the likelihood of the absence of vocals.
Liveness	Measures the probability of a live performance.
Valence	Describes the musical positiveness conveyed by the track.
Tempo	Estimated tempo expressed in beats per minute (BPM).

Unlike manually assigned genre labels, which reflect cultural conventions and historical evolution, these descriptors are directly extracted from the audio signal and therefore provide an objective representation of the acoustic properties of each song.

1.2 Data Standardization

Before constructing the unified corpus, both datasets required a preliminary standardization process to ensure a consistent representation of tracks, features, and genre labels.

The two sources were first aligned by removing unnecessary attributes and harmonizing their schema, retaining only the variables relevant to the objectives of this study. Records containing missing or inconsistent values in essential variables were discarded.

Finally, duplicate tracks appearing across both datasets were identified through metadata matching and removed to prevent the artificial weighting of repeated entries. When duplicates occurred, records from the second dataset were retained, as they provided more granular meta-data (as discussed in section 1.2.1).

1.2.1 Genre Harmonization

A secondary challenge concerned the semantic representation of musical genres. While numerical audio descriptors were directly comparable across both datasets, their underlying genre taxonomies differed substantially. The first dataset assigns each song to a relatively broad musical category, whereas the second employs a considerably more detailed classification composed of numerous specialized subgenres.

Directly combining these labels would have produced an inconsistent semantic representation, where broad categories coexist with highly specific stylistic labels describing different levels of abstraction. Such heterogeneity would distort both statistical analyses and the interpretation of the results.

To overcome this limitation, the original genre labels were harmonized into a unified set of **macro-genres**, while preserving the original subgenre metadata as an auxiliary attribute. This dual representation enables analysis at two complementary levels: macro-genres provide a consistent semantic baseline for global analyses, whereas the preserved subgenres allow for the investigation of fine-grained stylistic distinctions within each broader musical family.

1.3 Dataset Compatibility Assessment

Although the datasets had been standardized structurally, their integration still required validating that they represented compatible observations of the same underlying acoustic space. This verification is particularly important because the objective of this work is to discover intrinsic structures through unsupervised learning techniques. If systematic differences existed between the two data sources, clustering algorithms and graph-based analyses could identify dataset-specific artifacts rather than genuine musical relationships.

To evaluate the compatibility of the two datasets, the numerical Spotify descriptors shared by both collections were compared using the Kolmogorov–Smirnov (KS) test. For every analyzed feature, the null hypothesis that the two samples originated from the same underlying distribution was rejected (Table 1.2). This result is consistent with the markedly different composition of the two datasets rather than indicating an incompatibility between them. As shown in Table 1.3, the first dataset *D1* is primarily composed of a limited number of broad musical genres, whereas the second *D2* includes a considerably larger and more stylistically diverse collection, introducing numerous additional genres that are absent from the first dataset. Consequently, the merged corpus is expected to span a broader portion of the acoustic feature space, naturally leading to differences in the distributions of the numerical audio descriptors.

To further investigate whether these statistical differences reflected genuine inconsistencies or simply a broader musical coverage, the empirical distributions of the principal audio descriptors were visually inspected (Figures A.1 and A.2). Although differences between the datasets are evident, the overall shapes and value ranges remain comparable across all features.

Finally, a complementary validation was performed by comparing the correlation structures of the numerical descriptors. Figure 1.1 shows the correlation matrices computed independently for the two datasets. Despite the differences in genre composition and feature distributions, the principal relationships among the Spotify audio features remain remarkably consistent.

Table 1.2: KS Test Results

Feature	KS Stat	<i>p</i> -value
danceability	0.231	< 0.001
energy	0.141	< 0.001
loudness	0.147	< 0.001
speechiness	0.137	< 0.001
acousticness	0.208	< 0.001
instrumentalness	0.106	< 0.001
liveness	0.045	< 0.001
valence	0.098	< 0.001
tempo	0.086	< 0.001

Table 1.3: Macro Genre Composition

Genre	D1	D2	Genre	D1	D2
Alternative	0	3,365	Metal	0	6,035
Ambient/Chill	0	5,602	Mood	0	3,003
Classical	0	2,256	Pop	4,836	5,086
Electronic	4,603	15,065	Reggae	0	2,600
Folk/Country	0	4,492	Regional	0	5,499
Hip-Hop/R&B	9,232	1,365	Rock	3,739	4,650
Instrumental	0	868	Soundtrack	0	3,553
Jazz/Blues	0	2,189	Spoken	0	987
Kids	0	1,827	World	0	5,327
Latin	3,819	7,574			

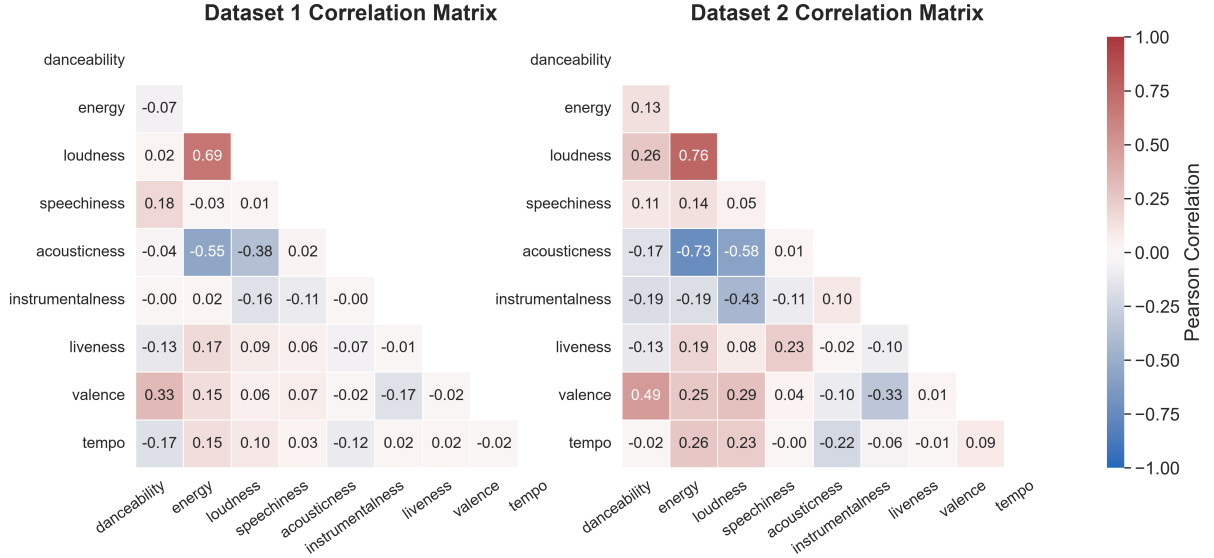


Figure 1.1: Correlation matrices of numerical audio descriptors computed independently for Dataset 1 (left) and Dataset 2 (right).

Although the two datasets exhibit differences in their feature distributions, their comparable value ranges and similar correlation structures suggest that they provide complementary coverage of the same set of acoustic descriptors.

1.4 Preliminary Exploratory Data Analysis

Following the compatibility assessment, the two standardized datasets were merged into a single collection that constitutes the basis for all subsequent analyses. The resulting dataset contains 106,008 unique songs described by 9 variables, combining the broader stylistic coverage of the larger dataset with the complementary information provided by the smaller one.

A preliminary exploratory analysis was then conducted to characterize the statistical properties of the integrated corpus before applying any unsupervised learning technique. The distribution of songs across the resulting macro-genres, shown in Figure 1.2, reveals a markedly unbalanced composition, with a limited number of genres accounting for the majority of the observations while several others are represented by comparatively fewer tracks.

To further characterize the acoustic space, the correlation matrix of the merged dataset was computed. Figure 1.3 illustrates the relationships among the Spotify audio descriptors after integration. Several moderate and strong correlations are observed, indicating that multiple musical characteristics frequently co-occur.

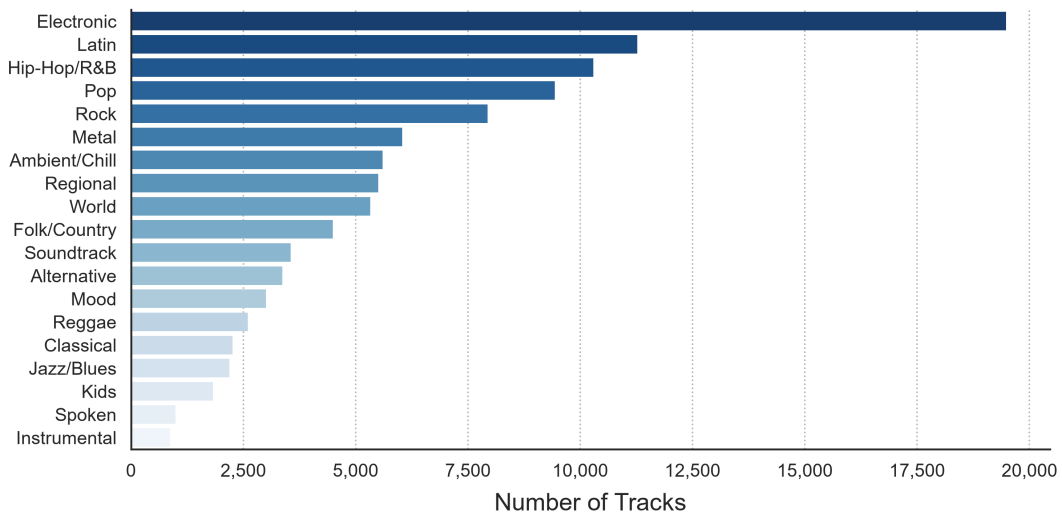


Figure 1.2: Distribution of tracks across macro-genres in the merged dataset.

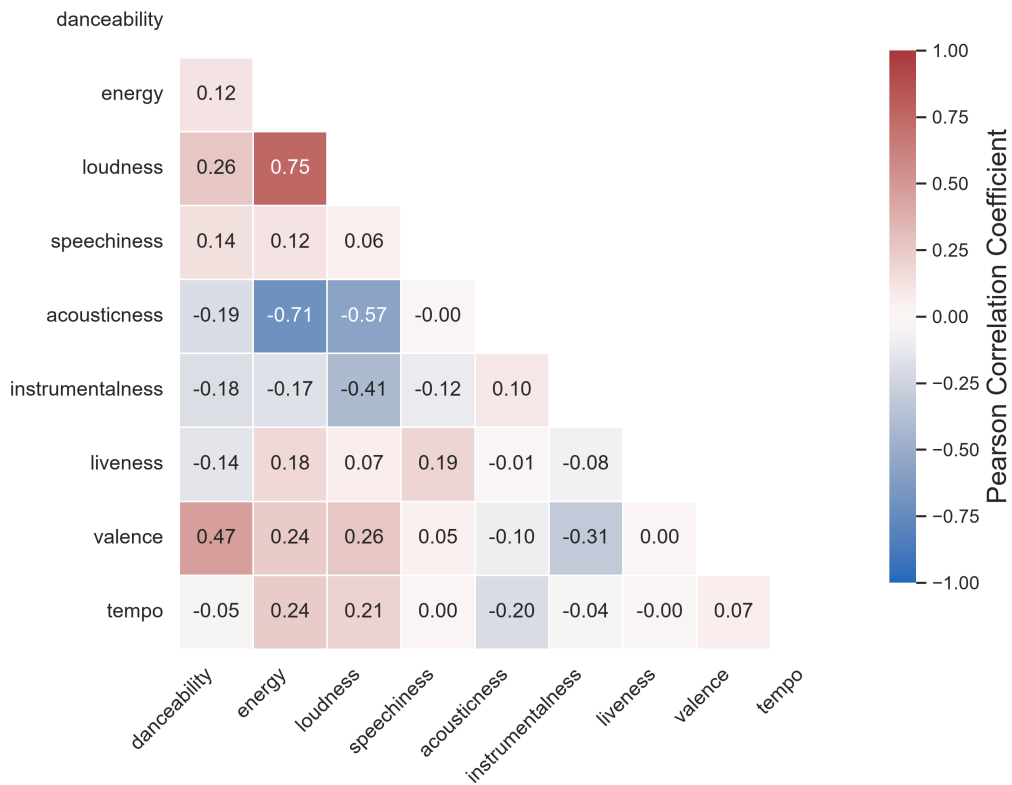


Figure 1.3: Correlation matrix of Spotify audio descriptors computed on the integrated dataset.

While this preliminary analysis provides an overview of the statistical properties of the data, it does not reveal whether traditional genre labels correspond to naturally emerging acoustic groups. Addressing this question requires unsupervised techniques capable of exploring the geometry of the feature space, which are introduced in the following chapter.

2 Can Musical Genres Be Recovered from Acoustic Features?

2.1 Motivation

The central question addressed in this work is whether traditional musical genres correspond to intrinsic structures of the acoustic feature space. If genre labels accurately describe the acoustic characteristics of songs, one would expect tracks belonging to the same genre to form compact and well-separated groups when represented through Spotify’s numerical audio descriptors.

Unsupervised clustering provides a natural framework for investigating this hypothesis, as it attempts to identify groups of similar observations without relying on any prior knowledge of the existing genre labels. A strong correspondence between the resulting clusters and the semantic genre annotations would therefore suggest that genres emerge naturally from the acoustic feature space. Conversely, poor agreement would indicate that the organization of music cannot be fully explained by traditional genre classifications.

To evaluate this hypothesis, K-Means clustering was applied to the cleaned dataset described in the previous chapter. Its results were subsequently compared with the original genre labels using both internal clustering metrics and external validation measures.

2.2 Acoustic Feature Representation

Each song was represented through the continuous Spotify audio descriptors introduced in Table 1.1. These variables constitute a compact numerical description of the acoustic characteristics of each track and therefore define the feature space used throughout the clustering analysis.

Prior to clustering, all continuous variables were standardized using **z-score normalization** to ensure that features measured on different numerical scales contributed equally to the distance calculations employed by the clustering algorithms.

Key and Mode Features Preliminary experiments were conducted both including and excluding the variables key and mode, with the musical key represented through one-hot encoding. Although these descriptors encode relevant musical information, they represent discrete harmonic characteristics rather than continuous acoustic properties. Their inclusion consistently reduced the quality of the resulting clusters as can be seen in Figure 2.1. Consequently, all subsequent clustering experiments were performed using only the standardized continuous audio descriptors.

Principal Component Analysis Principal Component Analysis (PCA) was employed to explore the geometry of the acoustic feature space and to provide a low-dimensional visualization of the clustering results. Preliminary experiments also evaluated clustering in the reduced PCA space. Although the resulting clustering quality was slightly higher, the improvement was marginal (Figure 2.1). Since the original feature space contains only nine continuous descriptors, dimensionality reduction would mainly reduce interpretability rather than computational complexity (see PCA loadings in Figure A.3). Consequently, PCA was retained exclusively as a visualization tool.

Figure 2.2 provides an initial qualitative assessment of the organization of the acoustic feature space. Most macro-genres occupy broad and highly overlapping regions, with no clear boundaries separating one genre from another. While certain genres exhibit a tendency to concentrate in specific areas of the projection, these regions remain largely interconnected and do not form distinct, isolated clusters. Only a small number of categories, such as Spoken and Instrumental,

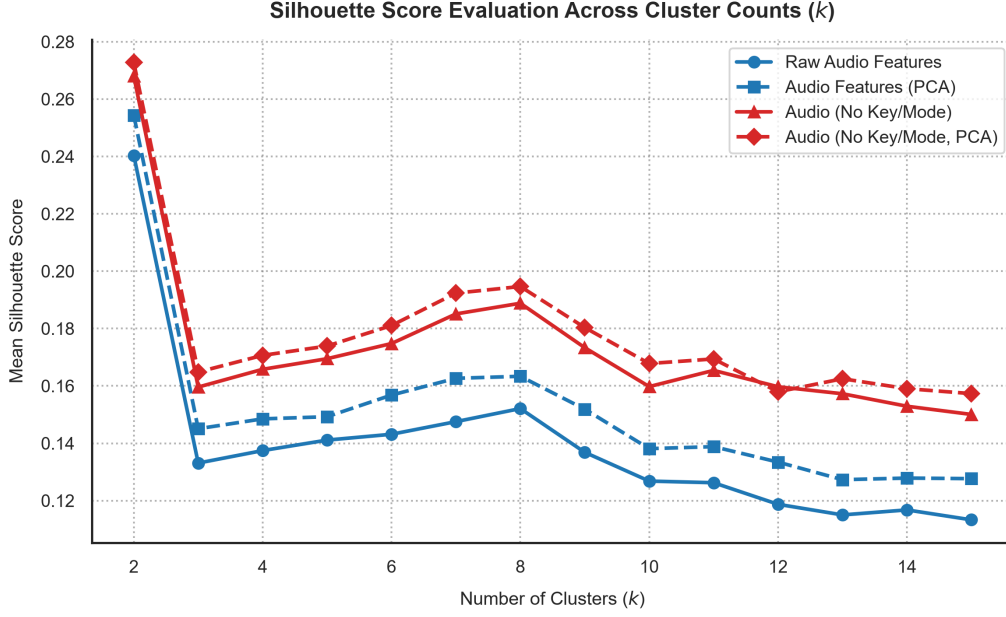


Figure 2.1: Comparison of mean Silhouette scores across varying numbers of clusters (k) for raw and PCA-transformed feature representations, with and without key/mode descriptors.

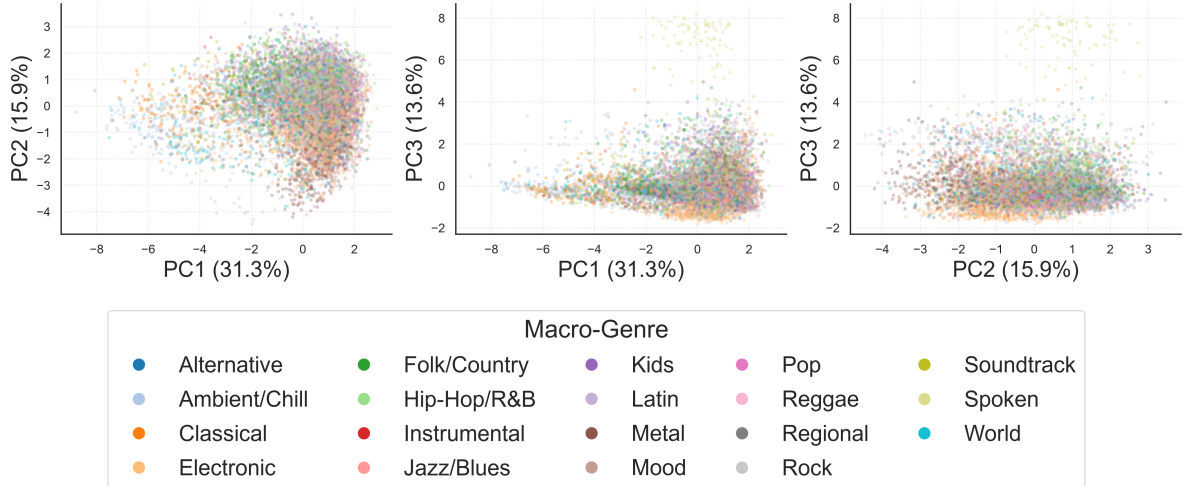


Figure 2.2: Pairwise PCA projections of the standardized acoustic feature space, colored by macro-genre. The first three principal components account for 60.8% of the total variance.

display a clear localization, reflecting their unique acoustic characteristics rather than a general property of musical genres. Overall, the PCA projection suggests that the acoustic feature space is better described as a continuous manifold with overlapping genre distributions than as a collection of well-separated genre-specific regions.

2.3 Clustering Methodology

To evaluate whether genres emerge naturally from the acoustic feature space, K-Means was adopted as the primary clustering baseline because it scales efficiently to the size of the dataset and produces centroid-based partitions that are straightforward to interpret. Its tendency to identify compact and approximately spherical groups makes it a useful baseline for assessing whether genre labels correspond to clearly separable acoustic regions.

Clustering quality was evaluated using both internal and external validation metrics. The Silhouette Score [6] was used to assess cluster compactness and separation independently of the genre labels, while the Adjusted Rand Index (ARI) [2] and Normalized Mutual Information (NMI) [7] measured agreement between the resulting clusters and the macro-genre taxonomy.

Parameter Selection Since K-Means requires the number of clusters k to be specified in advance, the algorithm was evaluated across a range of candidate values. The Silhouette Score was then used to identify values of k that provided an appropriate balance between within-cluster compactness and separation between clusters.

2.4 Results on Macro Genres

The clustering experiments were designed to investigate whether the acoustic feature space naturally partitions into groups corresponding to traditional musical genres. Rather than evaluating a single clustering configuration, three complementary experiments were performed. The first two explored the intrinsic organization of the acoustic space by selecting the number of clusters according to internal validation metrics, while the third explicitly evaluated whether enforcing a partition comparable to the number of macro-genres produced a stronger correspondence with the semantic labels.

2.4.1 Natural Acoustic Partitions

The Silhouette analysis showed a global maximum at $k = 2$ and a secondary local maximum around $k = 8$. These two configurations were therefore selected to examine the acoustic organization at two different levels of granularity.

Figure 2.3a presents the distribution of macro-genres across the two clusters obtained with $k = 2$. Rather than separating songs according to their semantic genre labels, the algorithm identifies two broad acoustic profiles that are shared across multiple genres. Most macro-genres exhibit a clear preference for one of the two clusters, with particularly strong tendencies observed for genres such as *Metal* and *Reggae*. However, these preferences are rarely exclusive. Nearly every macro-genre remains represented in both clusters, although in different proportions, indicating that songs belonging to the same genre often span multiple acoustic profiles.

Figure 2.3b shows that increasing the number of clusters provides a finer partition of the acoustic space without substantially changing the overall picture. Most genres remain distributed across multiple clusters, while each cluster still contains songs from several genres. Only the Spoken category forms a nearly isolated acoustic group.

The cluster centroids provide additional evidence for this interpretation. As illustrated in Figure 2.4, each cluster is defined by a particular configuration of audio features rather than by a single prevailing musical genre. For instance, some clusters correspond to tracks with a strong speech component, others to predominantly acoustic and instrumental pieces, and others to highly energetic and dance-focused tracks.

These qualitative observations are consistent with the quantitative evaluation shown in Table 2.1. The Silhouette Score indicates that the identified clusters possess a reasonable internal structure, yet the external validation metrics remain low, demonstrating only limited agreement between the discovered acoustic groups and the existing genre taxonomy.

Table 2.1: Clustering performance metrics across varying cluster counts ($k = 2, 8$).

Clusters (k)	Silhouette Score	ARI	NMI
2	0.2680	0.0336	0.0653
8	0.1888	0.0599	0.1402

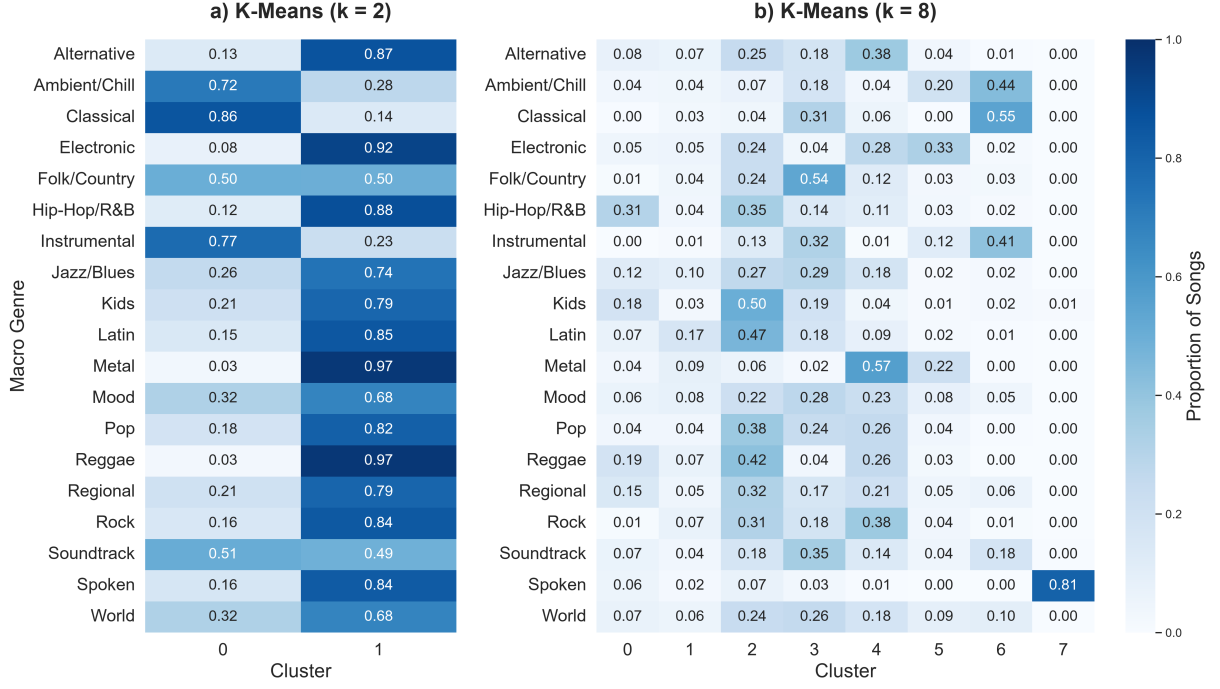


Figure 2.3: Distribution of macro-genres across the clusters obtained with K-Means for (a) $k = 2$ and (b) $k = 8$.

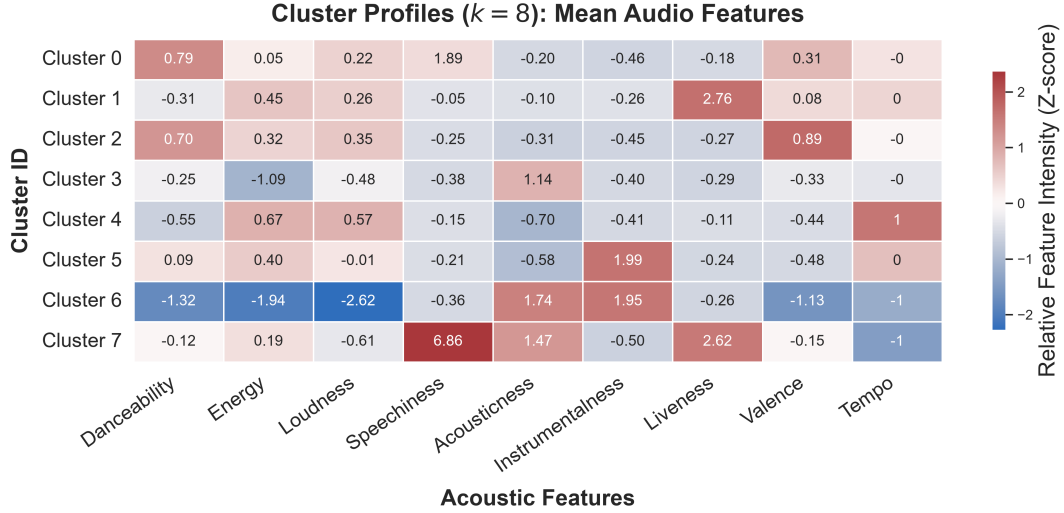


Figure 2.4: Average Audio Features by Cluster.

2.4.2 Genre-Oriented Partition

The previous experiments reveal the partitions that best describe the intrinsic organization of the acoustic feature space. However, they do not directly evaluate whether the predefined genre taxonomy itself can be recovered from the acoustic descriptors. To address this question, an additional experiment was conducted by fixing the number of clusters equal to the number of macro-genres present in the dataset ($k = 19$).

Figure 2.5 reports the row-normalized distribution of each macro-genre across the $k = 19$ acoustic clusters. If traditional musical genres mapped cleanly onto isolated, naturally separated regions of the acoustic feature space, one would expect each macro-genre to be predominantly concentrated within a single cluster. Instead, the opposite behavior is observed: nearly all macro-genres are broadly dispersed across numerous clusters, while individual clusters simultaneously

accommodate tracks spanning multiple semantic categories.

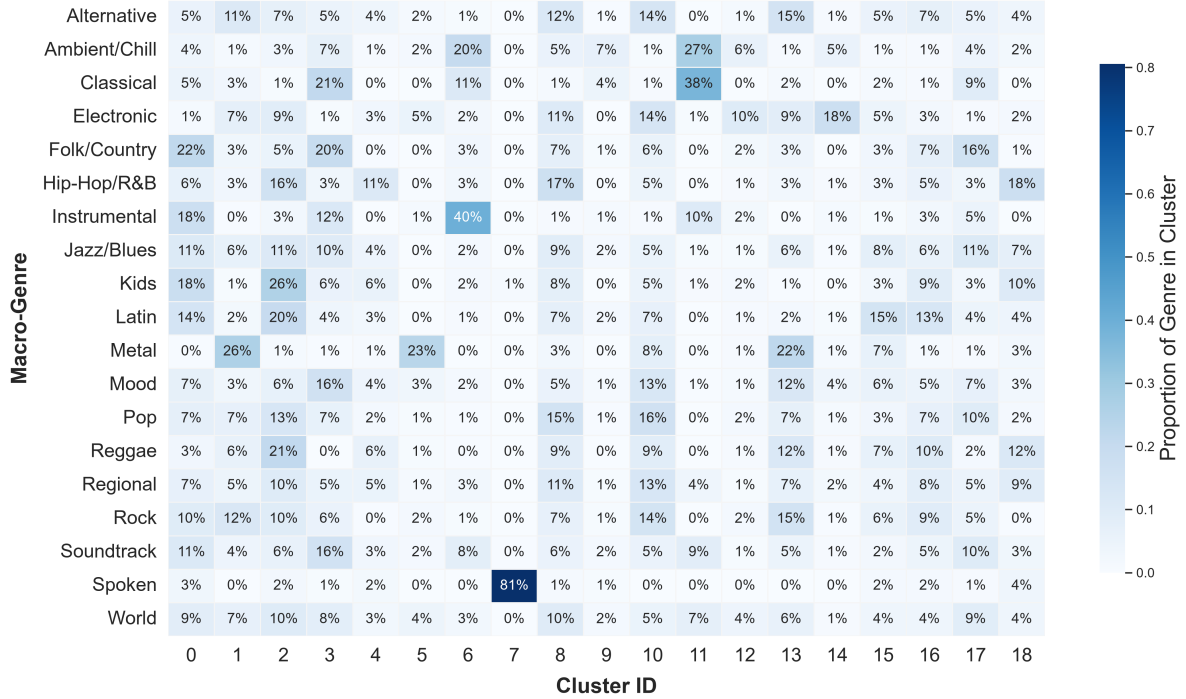


Figure 2.5: Cross-tabulation heatmap showing the row-normalized proportion of macro-genre tracks assigned across $k = 19$ clusters. Values indicate the percentage of a given genre residing in each cluster.

The external evaluation confirms this qualitative observation. Despite enforcing the number of clusters to match the number of semantic categories, the resulting partition achieves only $ARI = 0.0520$ and $NMI = 0.1412$, values comparable to those obtained in the previous experiments. Therefore, simply increasing the number of clusters or matching the expected number of genres does not improve the correspondence between acoustic similarity and semantic categorization.

2.5 Results on Subgenres

One possible explanation for the limited agreement observed at the macro-genre level is that these categories encompass a wide range of stylistically diverse music. To investigate whether finer semantic labels exhibit greater acoustic coherence, the clustering analysis was repeated independently for the three largest macro-genres: **Electronic**, **Latin**, and **Pop**. The analysis was restricted to songs for which granular subgenre annotations were available.

In this experiment, the number of clusters was matched to the number of semantic subgenres represented within each macro-genre, resulting in $k=20$ for Electronic, $k=9$ for Latin, and $k=6$ for Pop. This configuration provides a direct test of whether a partition containing the same number of groups as the semantic taxonomy is capable of recovering the corresponding subgenre structure. Figure 2.6 reports the distribution of the corresponding subgenres across the resulting clusters.

Despite matching the number of clusters to the number of semantic categories, the resulting partitions do not reconstruct the original subgenre taxonomy. This is particularly evident for Electronic, where most subgenres are distributed across numerous clusters rather than being associated with a single acoustic group. Latin exhibits a more heterogeneous pattern: some categories show relatively strong concentrations, most notably Tango and Reggaeton, whereas

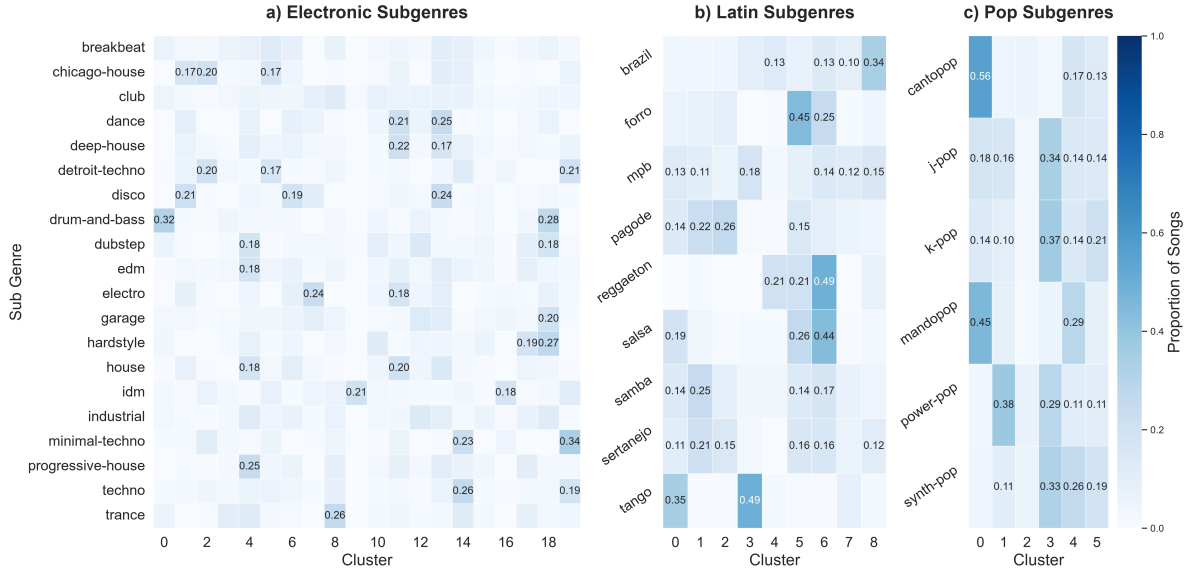


Figure 2.6: Row-normalized distribution of subgenres across K-Means clusters for Electronic, Latin, and Pop. For each macro-genre, the number of clusters was matched to the number of available subgenres. Only proportions above the annotation threshold are displayed for readability.

others, such as MPB, Samba, and Sertanejo, remain broadly distributed. A similar pattern is observed for Pop. Cantopop and Mandopop show comparatively strong associations with individual clusters, while J-pop, K-pop, Power Pop, and Synth-pop are distributed across several acoustic groups.

The subgenre analysis therefore provides a more nuanced picture than the macro-genre experiments. Certain stylistically distinctive categories exhibit stronger acoustic coherence, suggesting that some semantic distinctions are reflected in the audio descriptors. However, this behavior is not systematic: even when the number of clusters is explicitly matched to the number of subgenres, most categories do not map onto a unique acoustic cluster. Increasing semantic granularity therefore does not resolve the broader mismatch between the acoustic organization of the feature space and the existing genre taxonomy.

These results motivate a representation based on local acoustic similarity rather than discrete partitions, which is developed through a nearest-neighbour network in the following chapter.

3 Network Analysis

3.1 Motivation

The clustering results showed that acoustic similarity varies continuously across genre boundaries. To preserve the local relationships lost by hard partitioning, songs were therefore represented as nodes in a nearest-neighbour similarity network. Unlike clustering, a graph representation does not require sharp boundaries between groups. Instead, it explicitly preserves the local neighborhood of each song, allowing gradual acoustic transitions to remain visible.

3.2 Song Similarity Network

A song similarity network was constructed by representing each track as a node and connecting it to its k nearest neighbors in the standardized acoustic feature space.

Similarity was computed using **cosine distance**, which compares the relative acoustic profile of two songs rather than their absolute feature differences. Compared with Euclidean distance, cosine similarity produced networks with stronger community structure while maintaining comparable agreement with genre labels (Table A.1). Consequently, cosine distance was adopted for all subsequent analyses.

The neighborhood size was selected by evaluating several values of k using connectivity and community quality metrics. As shown in Table 3.1, the graph becomes fully connected at $k = 5$, while larger values primarily increase graph density and slightly reduce modularity without improving the correspondence between communities and genre labels. Therefore, $k = 5$ was selected as the smallest value yielding a connected network while preserving a well-defined community structure.

Table 3.1: Evaluation of different neighborhood sizes for the cosine-based k -nearest neighbor similarity network.

k	Connected Components	Clustering Coefficient	Communities	Modularity	ARI	NMI
3	2	0.216	38	0.872	0.046	0.135
5	1	0.238	26	0.843	0.061	0.135
8	1	0.252	22	0.821	0.067	0.136
10	1	0.257	21	0.812	0.059	0.132
15	1	0.268	19	0.797	0.064	0.130

3.3 Global Network Properties

The resulting similarity network contains 106,008 songs connected by 373,603 similarity edges. The graph forms a single connected component, indicating that every song can be reached from any other through acoustically similar tracks. This is consistent with the clustering results, supporting a continuous acoustic similarity space rather than isolated genre-specific regions.

As presented in Table 3.2, the average clustering coefficient (0.238) indicates locally cohesive neighborhoods, whereas the relatively low genre assortativity (0.193) shows that neighboring songs are only weakly constrained by genre labels, with local neighborhoods often containing songs from different genres despite similar acoustic characteristics.

Table 3.2: Summary statistics of the cosine-based song similarity network ($k = 5$).

Property	Value	Property	Value
Nodes	106,008	Edges	373,603
Average degree	7.05	Density	6.65×10^{-5}
Connected components	1	Largest component	100%
Average clustering coefficient	0.238	Genre assortativity	0.193

These structural properties motivate the application of community detection to identify the dense regions that naturally emerge within the similarity network. The coexistence of strong local cohesion and weak genre assortativity already suggests that acoustically meaningful neighborhoods extend beyond traditional genre boundaries.

3.4 Community Detection

To identify densely connected regions within the similarity network, community detection was performed using the Louvain algorithm [1]. The method partitions the graph by maximizing modularity, so that songs are more strongly connected within communities than across them.

The algorithm identified 26 communities with a modularity of 0.843. Because values close to one indicate much denser internal connectivity than expected in a random graph, this high modularity confirms a pronounced community structure despite the network’s global connectivity. Furthermore, community sizes were relatively balanced (Figure A.4), ranging from a minimum of 1,073 songs to a maximum of 6,979 (median: 4,074). Therefore, this structural organization is not artifactually driven by a few dominant groups or an abundance of tiny clusters.

These results show that the acoustic similarity network contains well-defined internal structure. However, modularity alone does not establish whether the detected communities correspond to traditional musical genres. This relationship is examined in the following section.

3.5 Communities versus Genres

To assess whether the network communities reproduce the predefined musical taxonomy, the Louvain assignments were compared with the macro-genre labels. The agreement was limited, with an Adjusted Rand Index of 0.061 and a Normalized Mutual Information of 0.135, indicating that the detected partition differs substantially from the genre classification.

Figure 3.1 examines this relationship in both directions. The first heatmap reports the genre composition of each community and shows that most communities contain songs from several macro-genres. The second reports how each genre is distributed across communities, revealing that individual genres are generally fragmented across multiple acoustic groups.

These results are consistent with the weak genre assortativity reported above. Although some communities display a stronger association with specific genres, the overall correspondence remains weak. Rather than reproducing semantic categories, the detected communities reveal an alternative organization based on acoustic similarity.

3.6 Acoustic Characterization of the Detected Communities

Having established the limited correspondence between communities and genre labels, their acoustic characteristics were examined through community centroids and representative song profiles.

Figure 3.2 presents the centroid of every detected community, computed as the mean stan-

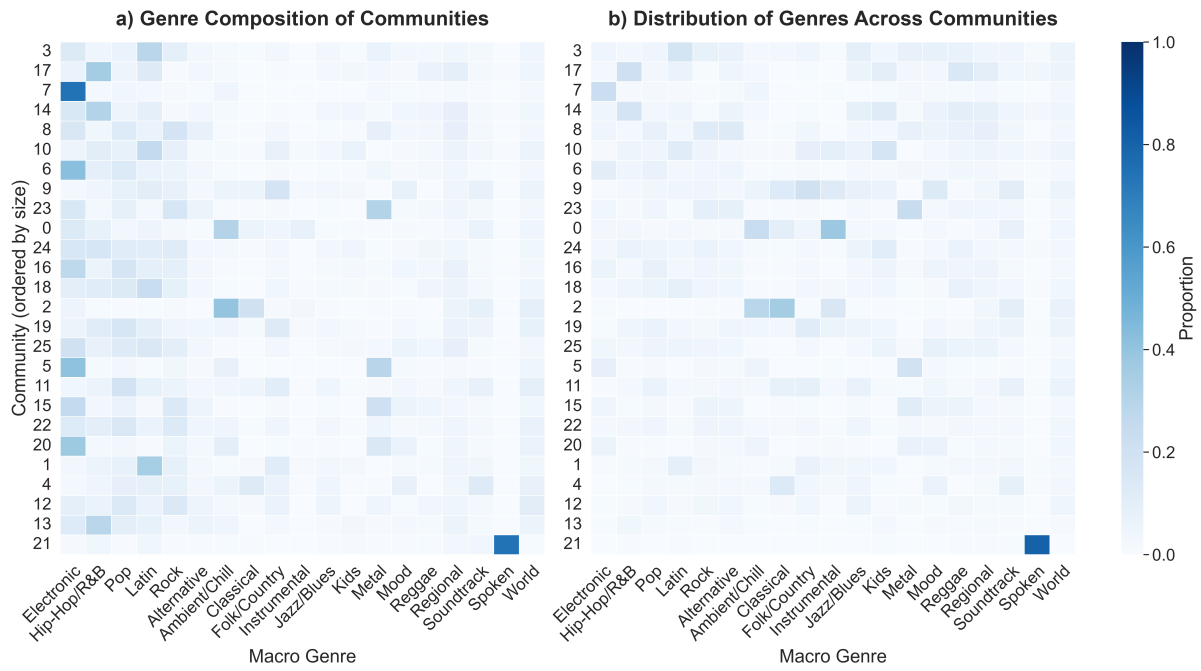


Figure 3.1: Relationship between Louvain communities and macro-genre labels in both directions. Communities are ordered by decreasing size.

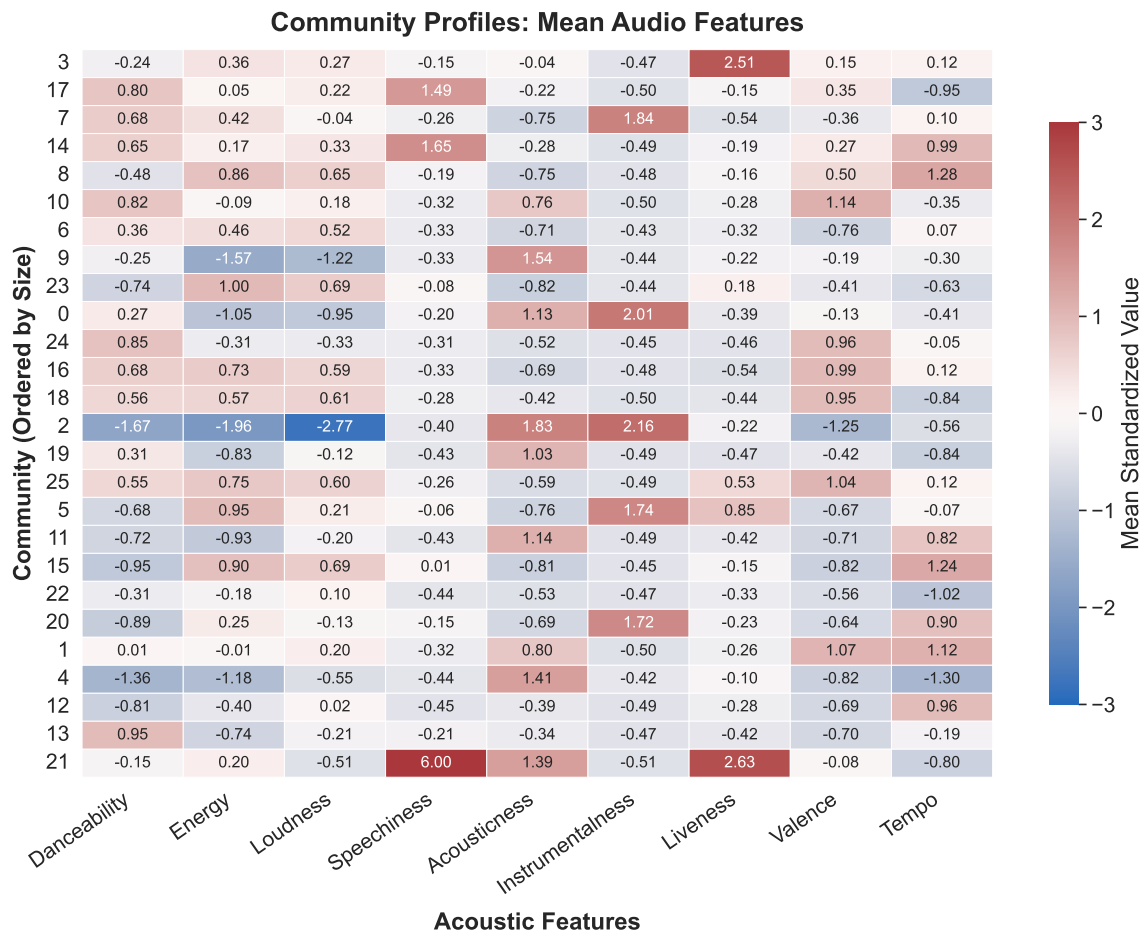


Figure 3.2: Average standardized audio features for each detected community.

standardized value of each audio feature. Although the communities were obtained exclusively from the topology of the similarity graph, rather than directly optimizing feature similarity, several distinctive acoustic profiles clearly emerge.

The detected communities span markedly different regions of the acoustic space. Some are characterized by dominant speech content, others by instrumental recordings or acoustic instrumentation, while others correspond to energetic, dance-oriented music. The emergence of these profiles despite the absence of genre information suggests that the detected communities capture coherent acoustic patterns rather than arbitrary graph partitions.

Although the average feature values provide an overview of each community, they do not indicate how consistently individual songs follow the corresponding community profile. To assess this aspect, Figure 3.3 reports representative radar plots for four communities together with randomly sampled songs belonging to each community. Since the features are standardized, the zero reference line represents the global dataset mean for each acoustic feature, allowing positive and negative deviations to be interpreted relative to the overall dataset.

Despite the variability observed among individual songs, the sampled tracks generally follow the characteristic profile of their respective community. Some features exhibit greater dispersion than others, indicating that the communities represent coherent local acoustic tendencies rather than uniformly compact groups. The centroids therefore provide useful summaries of the dominant acoustic characteristics of the selected communities without implying that all songs closely match the average profile.

To complement the visual characterization provided by the radar plots, within-community acoustic coherence was quantified by computing the cosine similarity between each song’s standardized audio-feature vector and the centroid of its assigned community. Higher values indicate that the acoustic profile of a song follows a direction similar to the average profile of its community.

Across all songs, the mean cosine similarity to the assigned community centroid was 0.792, with a median of 0.825 and an interquartile range of 0.719–0.899. At the community level, mean similarity ranged from 0.701 to 0.929 (Table A.2). Communities 21 and 2 exhibited the highest internal consistency, whereas even the lowest-scoring communities maintained mean similarities above 0.7.

Taken together, these quantitative results support the observations derived from the radar plots. Although individual songs display natural variation around the community centroid, they generally preserve the characteristic shape of the average acoustic profile. Consequently, the detected communities can be interpreted as acoustically coherent neighborhoods under the chosen feature representation.

3.7 Network Structure within Macro-Genres

To determine whether the limited correspondence between communities and genres was caused by the broad scope of the macro-genre taxonomy, the network analysis was repeated independently within selected macro-genres. In each subset, songs were connected according to acoustic similarity, Louvain communities were detected, and the resulting network structure was evaluated with respect to the corresponding subgenre labels.

Across all analyzed macro-genres, the same general pattern emerged. The detected communities exhibited substantially lower within-group acoustic dispersion than the original subgenres, indicating that the network partition identified groups of songs with more coherent acoustic profiles. At the same time, subgenre assortativity remained relatively low, ranging from 0.180 for Rock to 0.336 for Latin, suggesting that connections between acoustically similar songs were only weakly constrained by their semantic subgenre labels. This pattern was consistent across Electronic, Rock, Pop, and Latin.

The networks nevertheless displayed pronounced internal community structure. Modularity

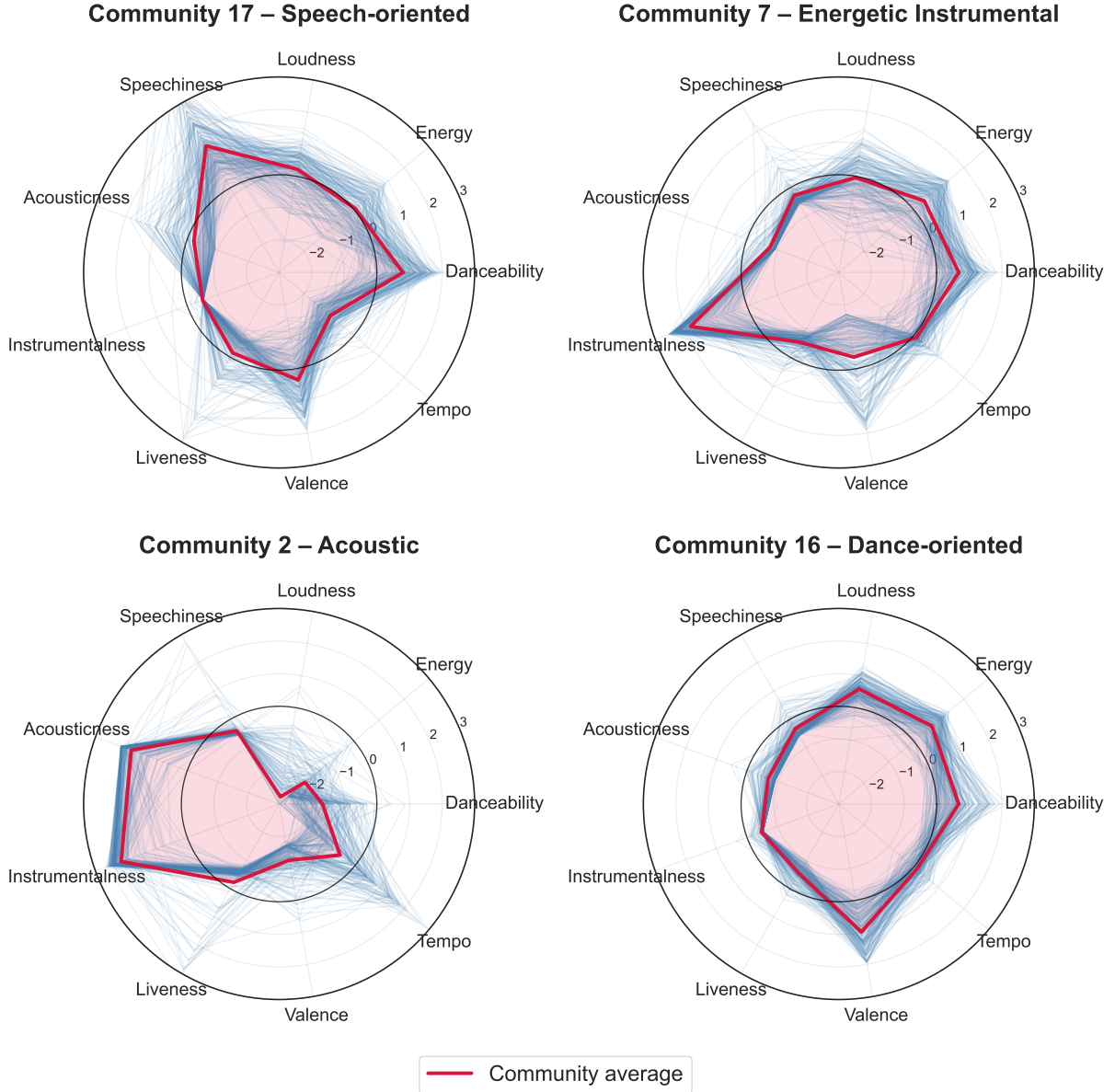


Figure 3.3: Representative acoustic profiles of selected communities. The red polygon represents the community centroid, the blue lines correspond to randomly sampled songs within each community, and the black circle indicates the global mean ($z = 0$) of the standardized features.

ranged from 0.786 to 0.820, while local consistency ranged from 0.828 to 0.860. Thus, even when the analysis was restricted to songs belonging to the same macro-genre, the similarity networks contained well-defined local acoustic structures that did not simply reproduce the existing subgenre taxonomy. This interpretation is further supported by the consistently lower cosine dispersion of the detected communities, which ranged from 0.171 to 0.200, compared with values between 0.423 and 0.520 for the original subgenres.

3.8 Structurally Central Songs within Communities

Having established that the detected communities remain mixed even when more granular subgenre labels are considered, the analysis was extended from the community level to the individual-song level. PageRank [5] was computed independently within each Louvain community to identify tracks occupying structurally central positions in the local similarity network.

Table 3.3: Network analysis within selected macro-genres. The table compares the structural properties of the subset similarity networks with the acoustic variability of the original subgenres and the detected communities.

Macro-genre	Subgenre assortativity	Modularity	Local consistency	Average within-group cosine dispersion	
				Subgenres	Communities
Electronic	0.199	0.820	0.860	0.423	0.176
Rock	0.180	0.791	0.828	0.520	0.171
Pop	0.258	0.786	0.828	0.520	0.200
Latin	0.336	0.797	0.841	0.438	0.175

Since the network is based on acoustic similarity rather than listener interactions, PageRank should not be interpreted as a measure of popularity but rather as an indicator of structural centrality within an acoustic neighborhood.

Across all communities, the PageRank percentile exhibited a positive but weak association with cosine similarity to the corresponding community centroid ($\rho = 0.182$). At the individual community level, the relationship remained consistently positive, with Spearman correlation coefficients ranging from approximately (0.11) to (0.30). A weaker negative association was observed between PageRank and Euclidean distance from the centroid ($\rho = -0.060$). These results suggest that structurally central songs tend to exhibit acoustic characteristics somewhat similar to those of their surrounding community, although the relationship remains weak.

A direct comparison between the highest-ranked song in each community and the corresponding community centroid revealed a more heterogeneous pattern. In 13 of the 26 communities, the highest-PageRank song exhibited a higher cosine similarity to the centroid than the average community member, whereas in the remaining 13 communities the opposite behavior was observed. This indicates that structural centrality does not necessarily coincide with proximity to the average acoustic profile.

These results show that structural centrality and centroid proximity capture different properties of a community. The centroid summarizes its average acoustic profile, whereas PageRank identifies songs that are central to its internal connectivity structure. The two measures therefore provide complementary descriptions of the detected communities

4 Discussion

4.1 Main Findings

The analyses conducted in this study consistently indicate that the organization of songs in the acoustic feature space only partially corresponds to the existing genre taxonomy. Both clustering and network-based methods identified meaningful acoustic structure, but neither reproduced macro-genre or subgenre labels with strong agreement.

The K-Means experiments showed that songs can be partitioned into acoustically interpretable groups, yet these groups remain semantically mixed. Increasing the number of clusters, including explicitly matching it to the number of macro-genres or subgenres, did not substantially improve the correspondence with the original labels. Some individual categories exhibited stronger acoustic concentration, but this behavior was not systematic across the taxonomy.

The network analysis provided complementary evidence. The cosine-based nearest-neighbour graph formed a single connected structure while exhibiting pronounced internal organization. Louvain detected 26 communities with a modularity of 0.843, but agreement with macro-genres remained limited, with $ARI = 0.061$, $NMI = 0.135$, and genre assortativity $= 0.193$. At the same time, the detected communities exhibited recognizable acoustic profiles and relatively high internal similarity, showing that the network partition captured meaningful acoustic organization even when it did not reproduce the semantic taxonomy.

The same pattern remained visible when the network analysis was repeated independently within selected macro-genres. In Electronic, Rock, Pop, and Latin, Louvain communities exhibited substantially lower within-group acoustic dispersion than the corresponding subgenres while maintaining only limited subgenre assortativity. Thus, increasing semantic granularity did not eliminate the mismatch between acoustic organization and genre labels.

4.2 K-Means and Louvain: Complementary Views of Acoustic Structure

K-Means and Louvain provide two different views of the same acoustic feature space. K-Means operates directly on the numerical descriptors and represents each cluster through a centroid, providing a compact global partition of the dataset. Louvain instead operates on the topology of the nearest-neighbour graph and identifies groups from patterns of local connectivity.

Despite these methodological differences, both approaches produced only limited agreement with genre labels. This consistency is important because it suggests that the weak correspondence observed with K-Means is not merely a consequence of selecting a centroid-based clustering algorithm. Replacing the global partition of the feature space with a network constructed from local acoustic similarities still produced groups that differed substantially from the semantic taxonomy.

The network representation nevertheless provides information that is not available from K-Means alone. Individual similarity relationships between songs are retained explicitly, allowing the analysis of local neighborhoods and the internal topology of the detected communities. This distinction became particularly visible through the PageRank analysis. Structurally central songs showed only a weak relationship with similarity to their community centroid, and the highest-ranked track was more similar to the centroid than the average community member in only 13 of the 26 communities. Structural centrality therefore does not simply identify the song closest to the average acoustic profile.

Centroids and PageRank consequently provide complementary descriptions of a community.

The centroid summarizes its average acoustic characteristics, whereas PageRank identifies tracks occupying central positions within its connectivity structure. Together, these results show that the network representation adds a song-level structural perspective to the global acoustic organization identified through clustering.

4.3 Implications for Music Discovery and Recommendation

The song similarity network also provides a possible practical use of the acoustic organization identified in this study. Because edges explicitly represent local acoustic similarity, the graph can be used to explore relationships between individual tracks without requiring them to share the same genre label.

One possible application is the construction of paths between songs. Given a source track s_0 and a target track s_m , the graph can provide a sequence

$$s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow s_m.$$

in which every consecutive pair is connected through acoustic similarity. By assigning each edge a cost equal to the cosine distance between its endpoints, a shortest-path algorithm can identify a route minimizing the cumulative acoustic distance between the two tracks.

For a path P , the local continuity of the sequence can be summarized through the mean cosine similarity between consecutive tracks:

$$\text{Smoothness}(P) = \frac{1}{m} \sum_{i=0}^{m-1} \text{cosine_similarity}(s_i, s_{i+1}).$$

This should not be interpreted as a perceptual measure of playlist quality. It provides only an acoustic criterion derived from the same feature representation used to construct the network. Nevertheless, it offers a direct way of illustrating how the topology identified in the analysis can be used to move between different regions of the acoustic space.

The resulting paths can be adjusted by modifying the edge-cost function. In particular, adding a fixed penalty for each hop favors shorter routes by balancing acoustic continuity against path length. The cost function could also incorporate additional criteria, such as track popularity or user preferences, extending the same framework toward more practical recommendation scenarios.

4.3.1 Illustrative Song-to-Song Paths

To provide a concrete example of this mechanism, two song-to-song paths can be examined using the similarity network by minimizing the cumulative cosine distance between consecutive tracks. The first connects songs belonging to different macro-genres, while the second connects songs sharing the same macro-genre but located in different acoustic regions.

The cross-genre example is intended to illustrate whether locally similar transitions can connect songs located on opposite sides of a semantic genre boundary. The within-genre example instead illustrates whether two tracks sharing the same broad semantic label may nevertheless occupy sufficiently different acoustic regions to require intermediate transitions in the similarity graph.

These examples should be regarded as demonstrations of the structure already identified in the network rather than as an evaluation of a complete recommendation system. Their purpose is to show how the local relationships preserved by the graph can be translated into an interpretable song-level exploration mechanism.

Table 4.1: Illustrative song-to-song paths in the acoustic similarity network.

Step	Track	Artist	Macro-genre	Comm.	Similarity to next
Cross-genre path					
1	Bohemian Rhapsody	Queen	Rock	12	0.969
2	Way to Fall	Starsailor	Classical	12	0.962
3	Something - Remastered 2009	The Beatles	Regional	12	0.953
4	Santa Claus Is Comin' To Town - 1991 Remix	The Beach Boys	Rock	24	0.930
5	Strangers (Remastered)	The Kinks	Regional	12	0.930
6	Abre Essas Pernas Pra Mim	Velhas Virgens	Alternative	24	0.932
7	Glass Onion - Remastered 2009	The Beatles	Rock	24	0.955
8	Everything I Touch Turns to Tears	The Coffinshakers	Alternative	24	0.965
9	Espejismo	La Tropa Loca	Rock	24	0.978
10	3 small words	MisterWives	Alternative	24	0.981
11	I'm Missin' You Babe	Lebrado	Jazz/Blues	24	0.984
12	Un autre que moi	Fishbach	Pop	18	0.992
13	Touch Your Spot	Tucka	Jazz/Blues	18	0.991
14	Nira Nirayayi	Riya Das; Anaida Ancil; Sariga	World	18	0.991
15	Low	Lenny Kravitz	Rock	18	0.987
16	Shape of You - Galantis Remix	Ed Sheeran	Electronic	18	—
Within-genre path					
1	Shape of You - Galantis Remix	Ed Sheeran	Electronic	18	0.987
2	Tres Gotas De Agua Bendita	Gloria Estefan	Latin	18	0.990
3	El Mismo Sol - Radio Edit	Alvaro Soler; Don Lore V	Latin	16	0.993
4	MENEITO	Nil Moliner; Yera	Regional	16	0.981
5	Mind Blown	Syn Cole	Electronic	16	0.979
6	It's Sheryl Amazing	Terrence Parker	Electronic	16	0.976
7	Tenderness	Orlando Voorn	Electronic	7	0.985
8	Bac N Da Day - Clepto's Mix	Frankie Knuckles	Electronic	7	0.982
9	In Da Jungle - Elliot Fitch Remix	Orlando Voorn; Elliot Fitch	Electronic	7	0.982
10	Crazy Things at Night	Sonny Fodera	Electronic	7	0.974
11	Jack Yo Body	Harry Romero	Electronic	7	0.970
12	Hooked	Martin Ikin	Electronic	7	0.991
13	Losing It	FISHER	Electronic	7	—

Conclusions

This study examined the relationship between semantic genre classification and the organization of songs in an acoustic feature space. The results show that the two are related, but do not generally define the same structure.

Both the clustering and network analyses identified meaningful acoustic organization, while exhibiting only limited agreement with macro-genre and subgenre labels. This convergence across methods indicates that the observed mismatch cannot be attributed solely to the assumptions of a particular algorithm. Instead, songs belonging to the same genre frequently occupy different acoustic regions, while acoustically similar tracks may cross conventional genre boundaries.

The research question can therefore be answered by distinguishing between semantic genre labels and the acoustic organization considered in this study. Genres do not generally correspond to isolated and internally homogeneous regions of the selected feature space, while acoustic similarity provides a complementary organization based on continuous profiles and local relationships between songs.

The network representation makes this complementary structure directly accessible. By preserving song-level similarity relationships, it can support the identification of structurally central tracks and the construction of gradual transitions between acoustically different songs. These possibilities should be regarded as extensions of, rather than substitutes for, recommendation approaches incorporating listener behavior, semantic information and perceptual feedback.

Bibliography

- [1] Vincent D. Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008. doi:[10.1088/1742-5468/2008/10/P10008](https://doi.org/10.1088/1742-5468/2008/10/P10008).
- [2] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985. doi:[10.1007/BF01908075](https://doi.org/10.1007/BF01908075).
- [3] JoeBeachCapital. 30,000 spotify songs (kaggle dataset). Kaggle, n.d. Accessed: June 2026. URL: <https://www.kaggle.com/datasets/joebeachcapital/30000-spotify-songs/data>.
- [4] Maharshi Pandya. Spotify tracks dataset (kaggle dataset). Kaggle, n.d. Accessed: June 2026. URL: <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>.
- [5] Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. The pagerank citation ranking: Bringing order to the web. Technical report, Stanford InfoLab, 1999.
- [6] Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. doi:[10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7).
- [7] Nguyen Xuan Vinh, Julien Epps, and James Bailey. Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance. *Journal of Machine Learning Research*, 11:2837–2854, 2010.

Appendix

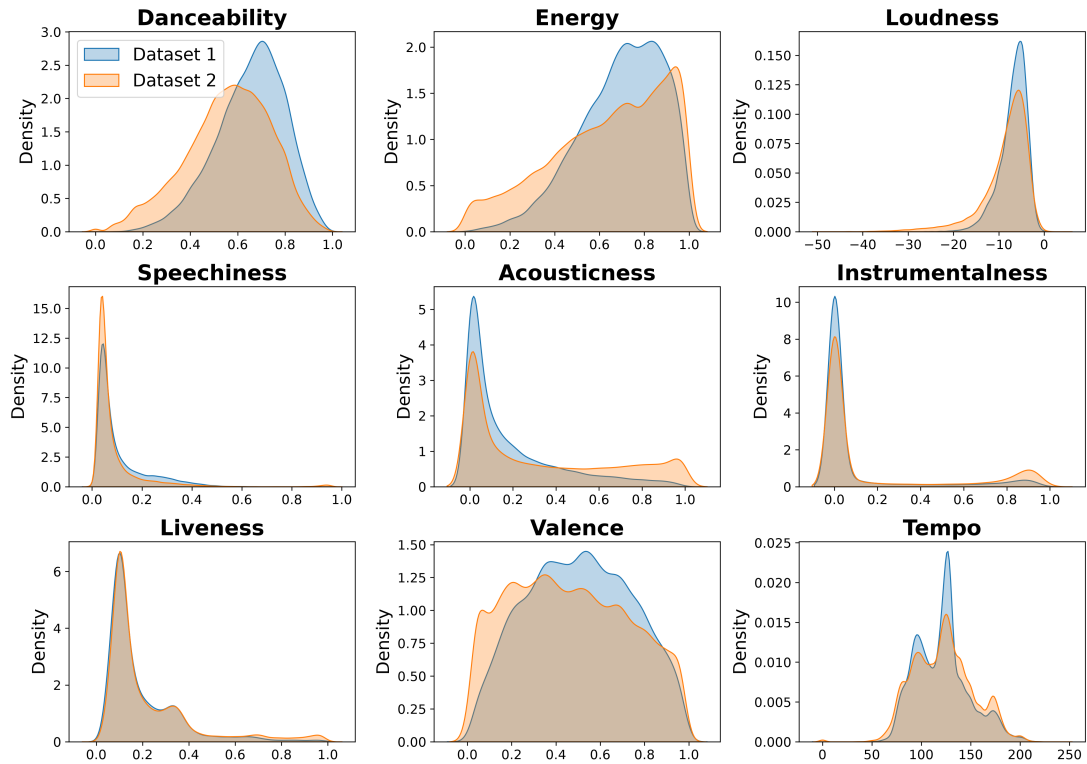


Figure A.1: Kernel Density Estimation (KDE) distributions comparing the 9 continuous audio features across Dataset 1 and Dataset 2.

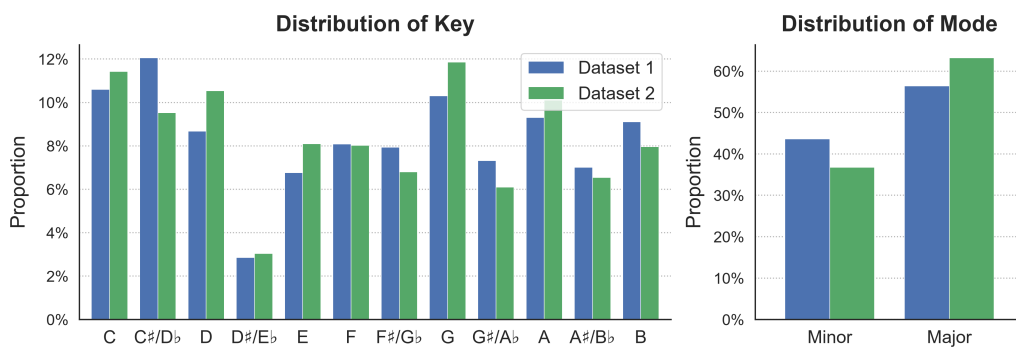


Figure A.2: Normalized distribution comparison for musical key and mode across both datasets.

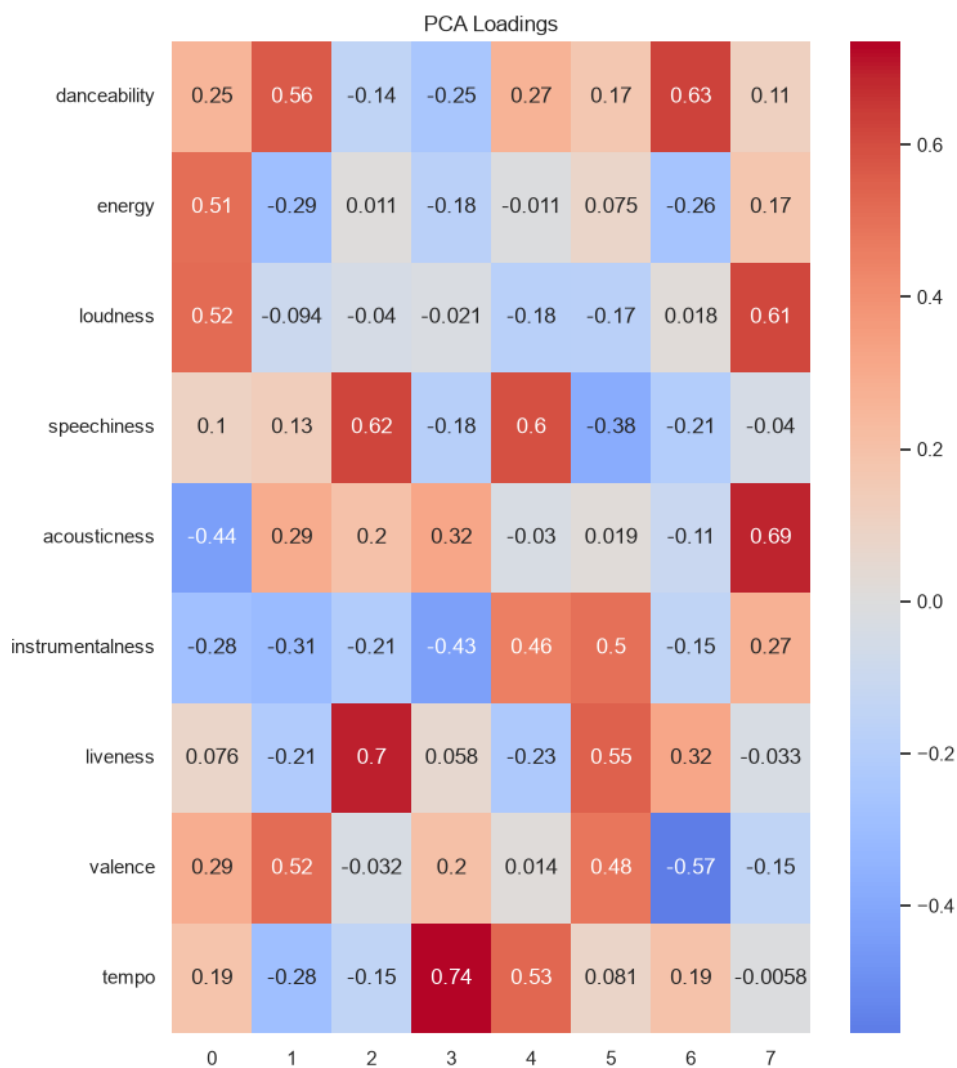


Figure A.3: PCA Factor Loadings

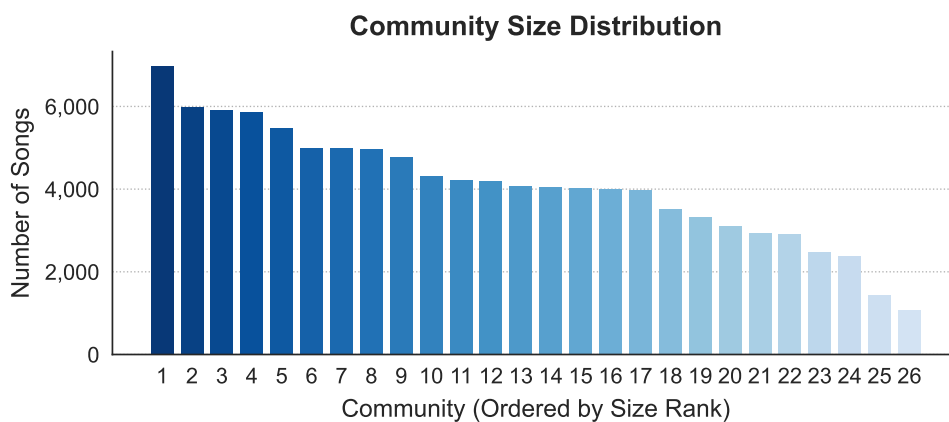


Figure A.4: Distribution of the number of songs across the 26 communities detected by the Louvain algorithm.

Table A.1: Comparison of Euclidean and cosine distance for different neighborhood sizes.

k	Distance	Clustering Coefficient	Modularity	Local Community Consistency	Genre Assortativity	ARI	NMI
5	Euclidean	0.151	0.835	0.868	0.194	0.070	0.136
5	Cosine	0.238	0.843	0.878	0.193	0.061	0.135
8	Euclidean	0.157	0.810	0.852	0.189	0.070	0.137
8	Cosine	0.252	0.821	0.862	0.190	0.067	0.136

Table A.2: Acoustic coherence of the detected communities.

Community	Songs	Mean cosine similarity	Std. dev.
21	1073	0.929	0.090
2	4063	0.905	0.106
16	4194	0.874	0.083
4	2474	0.866	0.122
15	3329	0.865	0.087
18	4085	0.841	0.115
8	5471	0.834	0.099
9	4975	0.834	0.119
11	3523	0.819	0.137
25	3999	0.816	0.114
19	4026	0.800	0.134
13	1442	0.794	0.104
23	4778	0.792	0.112
7	5912	0.789	0.147
6	4985	0.777	0.133
12	2379	0.776	0.140
10	5001	0.769	0.139
3	6979	0.763	0.146
0	4309	0.760	0.171
1	2924	0.750	0.148
24	4231	0.745	0.153
17	5997	0.738	0.152
5	3970	0.738	0.159
14	5855	0.735	0.141
22	3104	0.721	0.183
20	2930	0.701	0.165